July 22, 2025

Architecture Review

Initiative Name:	**I & O ExaData in Azure**			
OCIO Intake #				
Architecture Intake #	290			
Presentation Dates:	Candidate Architecture	07-21-2025	End-to-End Sketch	07-21-2025

canada life

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Initiative Overview

Initiative Description

- Oracle Exadata Hardware in the Canada Life environment will be past its support agreement on October 25, 2025. Currently, the solution is hybrid with the infrastructure on premise managed remotely by Oracle (OCI).
- A new strategic initiative has been established where the ExaData solution will be cloud based hosted in Azure (Exa@azure). The support agreement is now extended by one year to October 2026 by when the cloud-based solution will be implemented.

Note: *In Canada, Oracle Exadata is only available through Microsoft Azure. AWS offers the service, but not in this region.*

Business Unit	Infrastructure and Operations
Business Sponsor	Joe Kitchen/Hani Musa
Enterprise Architect	Goran Zugic
Portfolio Architect	Santhosh Kumar C, Ali Shamaei for Data
Delivery Lead	Glen L, Irene P, Bassem M
Project Level	L2
Governance	ARC
Timeline	Q2 FY25 – Q4 FY25

Budget

Infrastructure (1.2 M credits from Azure)	-1.15M
Vendor	1.7M
IT	.33M
Total	880K

Business Goals

- The proposed migration is part of Technology's broader modernization strategy to leverage cloud capabilities and improve our IT infrastructure.
- Currently, Oracle Exadata runs in the Canada Life Winnipeg/Scarborough Data Centres. The Oracle managed physical hardware Exadata runs on is due for mandatory refresh. Canada Life will migrate Exadata to the cloud vs refresh on premises, to align with the modernisation strategy.

Technology Goals

- Setup ExaData@Azure Infrastructure (in Azure)
- Test all Applications/Database(s)
- Productionalize all Applications (in WDC/SDC) and databases; those with performance issues to become part of ARM
- Decommission older hardware
- Development of all required artifacts in support of this initiative

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Initiative Overview (Workstream 1)

This project will be done in two phases called as Workstream 1 and Workstream 2. This document focuses exclusively on **Workstream 1**, which includes the following key activities:

1. Infrastructure Setup (Non-Production)

1. Provision the non-production environment for Exa@Azure.
2. Establish connectivity between on-premises applications and Exa@Azure.
3. The security implications of using data on Exa@Azure is being worked upon.

2. Performance Metrics Alignment

1. In later phases of Workstream 1, collaborate with app owners to get ready with application testing.
2. The goal is to identify key performance indicators and success criteria for evaluating system behavior.

3. Database Migration and Testing

1. Migrate a copy of the ExaCC (non-prod/prod environment) to Exa@Azure.
2. Connect the corresponding non-production applications to the new environment.
3. Engage application owners to perform testing and document their observations.

4. Decision Point

1. Once sufficient performance data is collected, the **initiative owner** will review the findings to determine the next steps.
2. The decision may involve either:
    1. Leave the applications on-prem and move Exadata to Azure, or
    2. Migrating applications to the Azure cloud, or
    3. Upgrading the existing **ExaCC** infrastructure.
3. The target date for this decision is **November 30, 2025**. Based on the outcome, **Workstream 2** will be planned to support the chosen direction.

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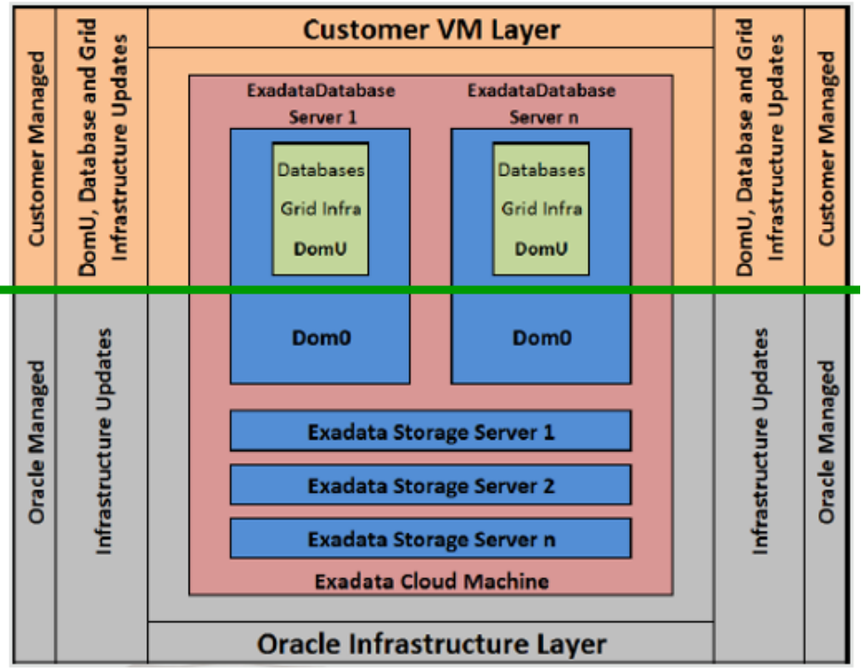
System Context

The Oracle Exadata Cloud at Customer platform is a self-contained database ecosystem that combines compute, storage, recovery, resiliency, monitoring and management within a managed solution.

- Canada Life manages VMs for databases
  - Only CL has credentials to access CL VMs and databases, can leverage Oracle Automation
- Canada Life configures and manages databases
  - Manage database schemas, users, database access and encryption keys

- Oracle owns and manages Exadata infrastructure
  - Servers, storage, storage software, networking inside rack, firmware, hypervisor, etc.



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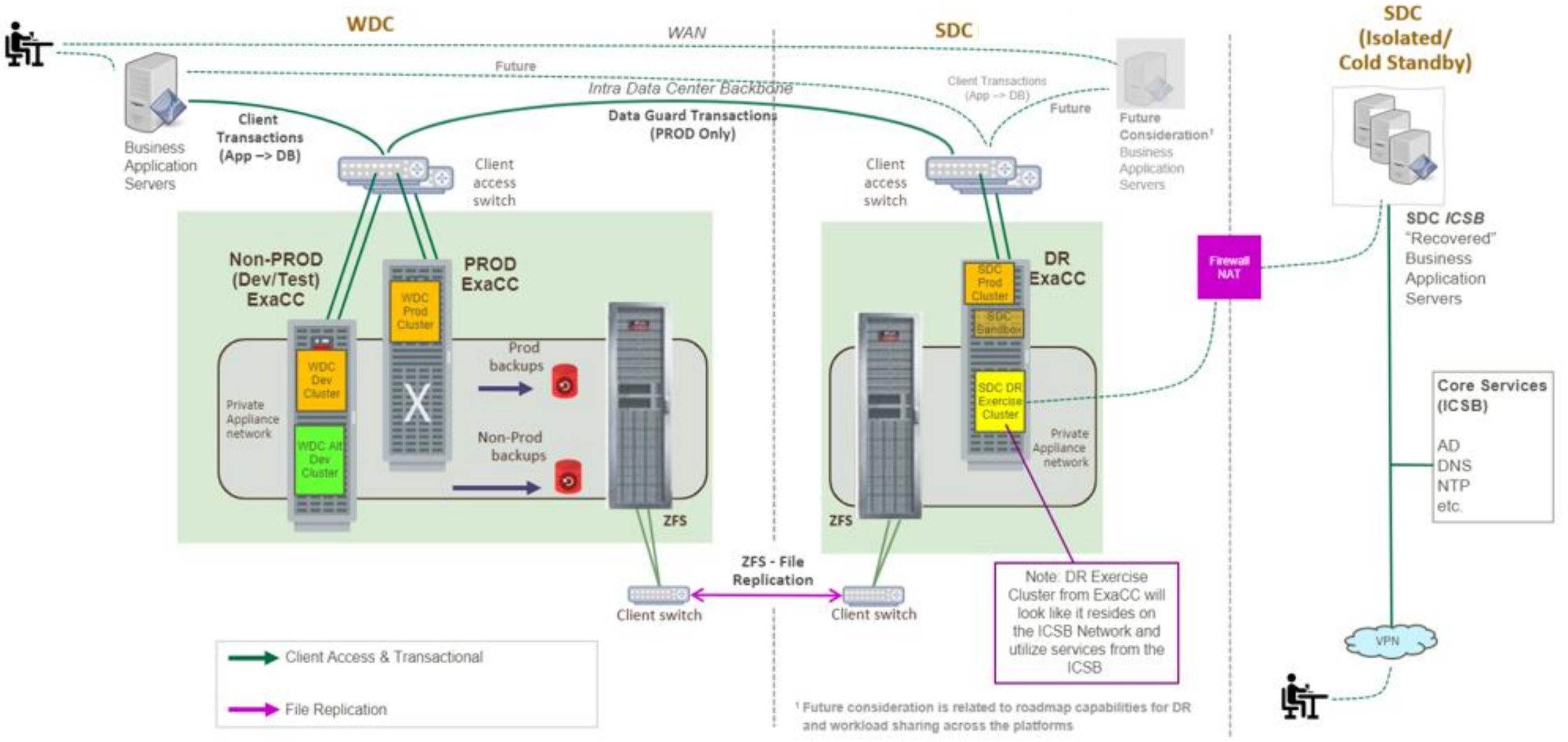
Physical Architecture



Note: OCI has secure network access to Exa@CC (not shown in diagram).

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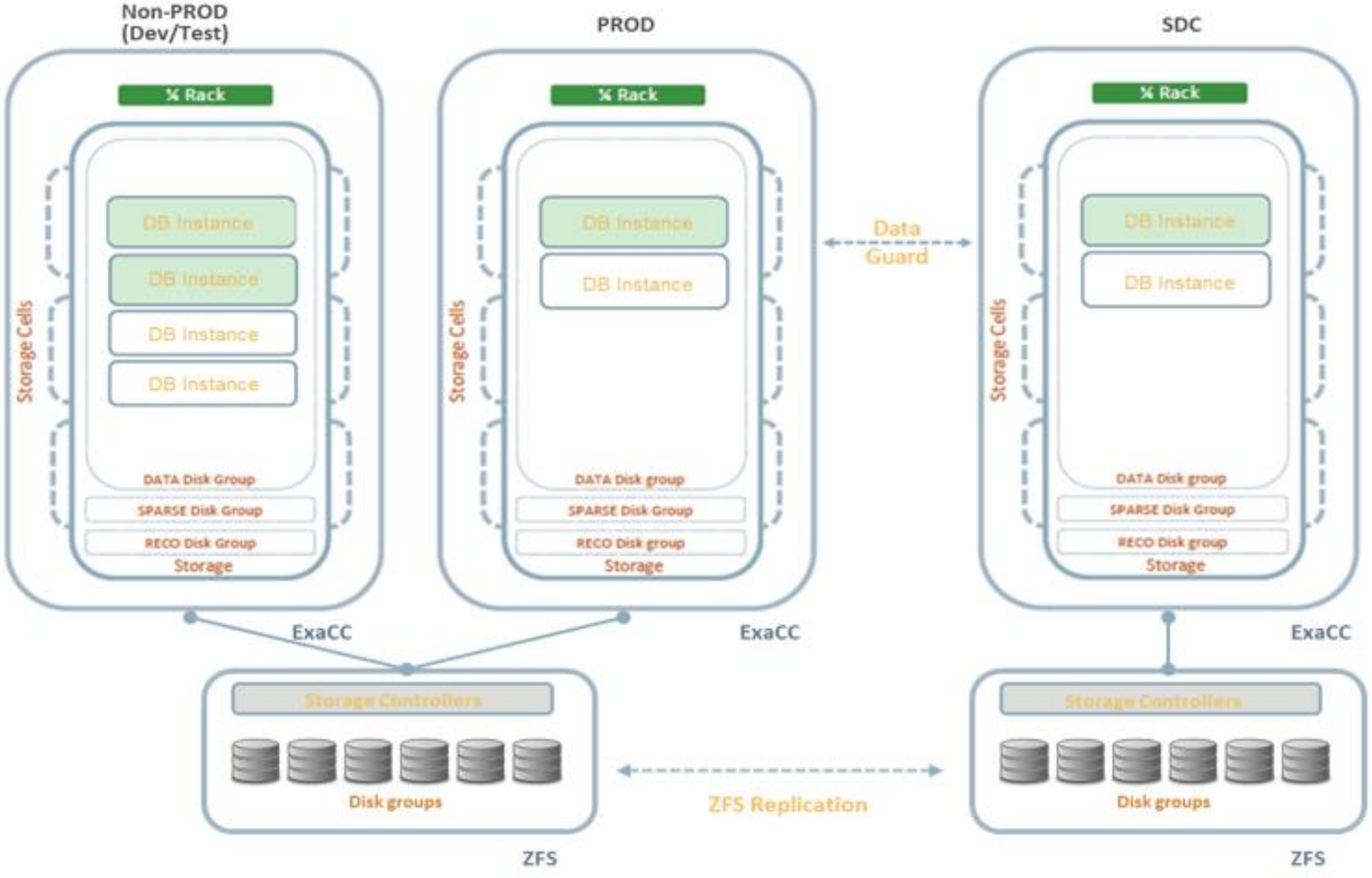
Infrastructure Detail

ExaData Configuration	
Compute Nodes (sometime referred to as Database Servers)	2
Max Number of OCPUs	100
Total Memory	1440 GB
No. of Storage Servers	3
Total Usable Internal Disk Capacity	150 TB

ZFS Configuration	
Initial Disk Capacity	700 TB
Expansion Capability	14PB



Solution Characteristics

Is this solution aligned to an approved pattern?	**Yes (some customisation might be required)**
Is this solution aligned to an approved Reference Architecture?	Yes (some customisation might be required)
Where will this solution be hosted?	❑ SaaS ❑ Public Cloud (IaaS, PaaS) ❑ Private Cloud (IaaS, PaaS) ❑ On-Prem ❑ Combination (highlight all that apply)
Will this solution introduce a new product into the IT standards?	No
What is the scope of the solution?	❑ Single Division/Function/Business Unit ❑ Multiple Divisions/Functions/Business Units ❑ Enterprise
What are the dependencies/interfaces for this solution?	❑ No change ❑ New/changed Internal interfaces ❑ External (3rd Party) System Dependency/Interface ❑ New Enterprise Shared Service(s)

Which experiences will be impacted?	❑ Customer ❑ Advisor ❑ Sponsor ❑ Employee
Will this solution handle PII data?	Yes
Which Applications are impacted? (include APM codes)	All 42 applications that are using ExaCC. Microsoft Excel Worksheet

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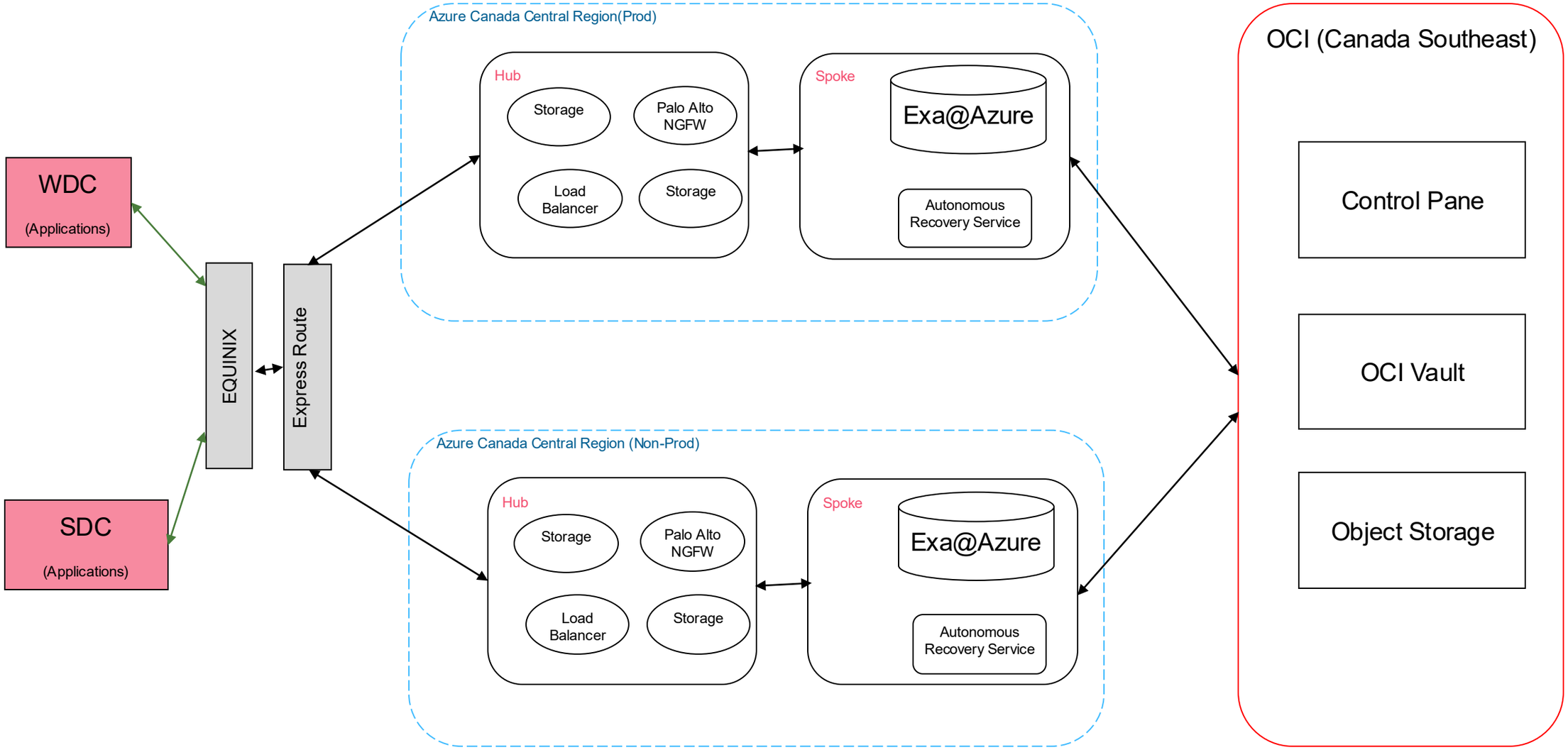
Logical Architecture



*Applications are in SDC and WDS data centers

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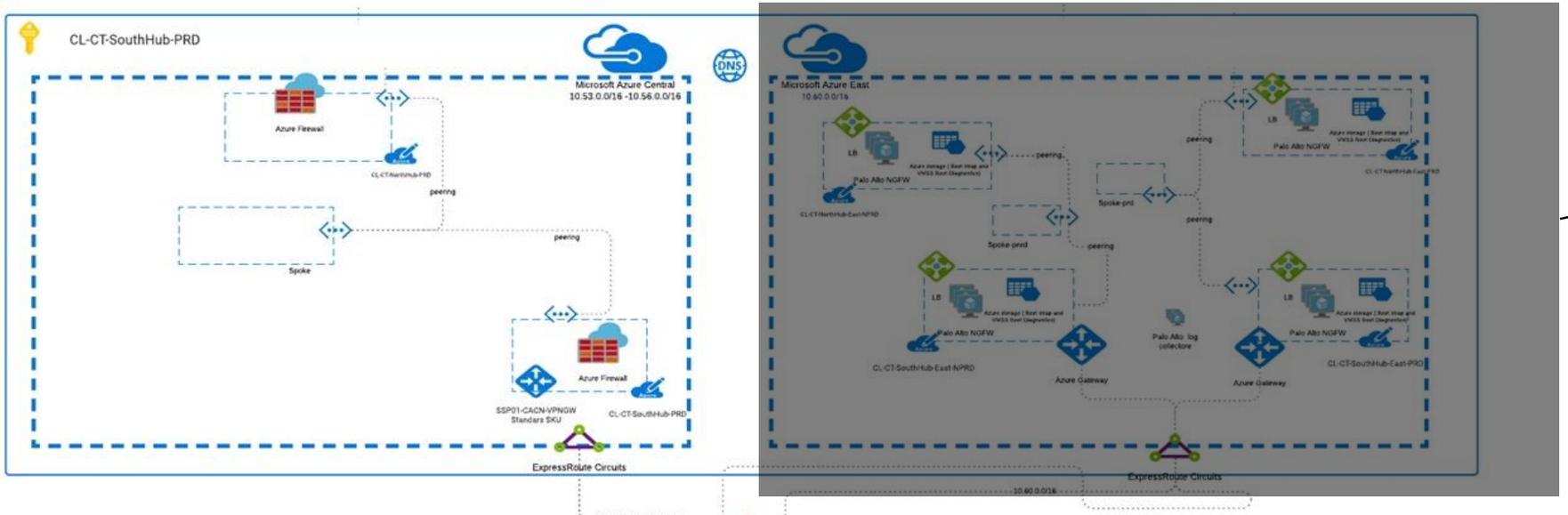
Design Pattern CL (System Context)

Azure Canada East/Central



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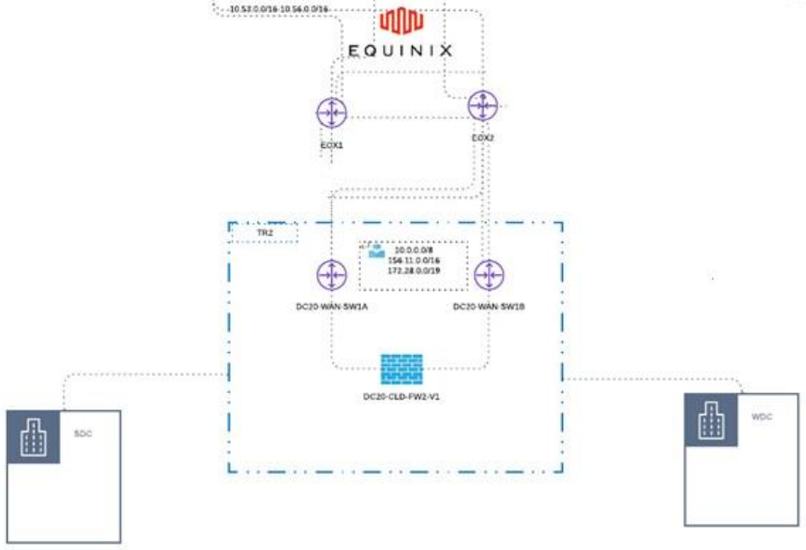

Reference: [AZ - Equinix Connections - Cloud & Automation - PRD-CONFLUENCE]

The image here shows established pattern used in Canada Life for MS Azure Connectivity with CL Datacenters.

Next slide shows detail view of Azure Canada Central region.

Disclaimer: *The image used above is sourced from the referenced material. It is assumed that the information depicted is accurate and exists as presented.*

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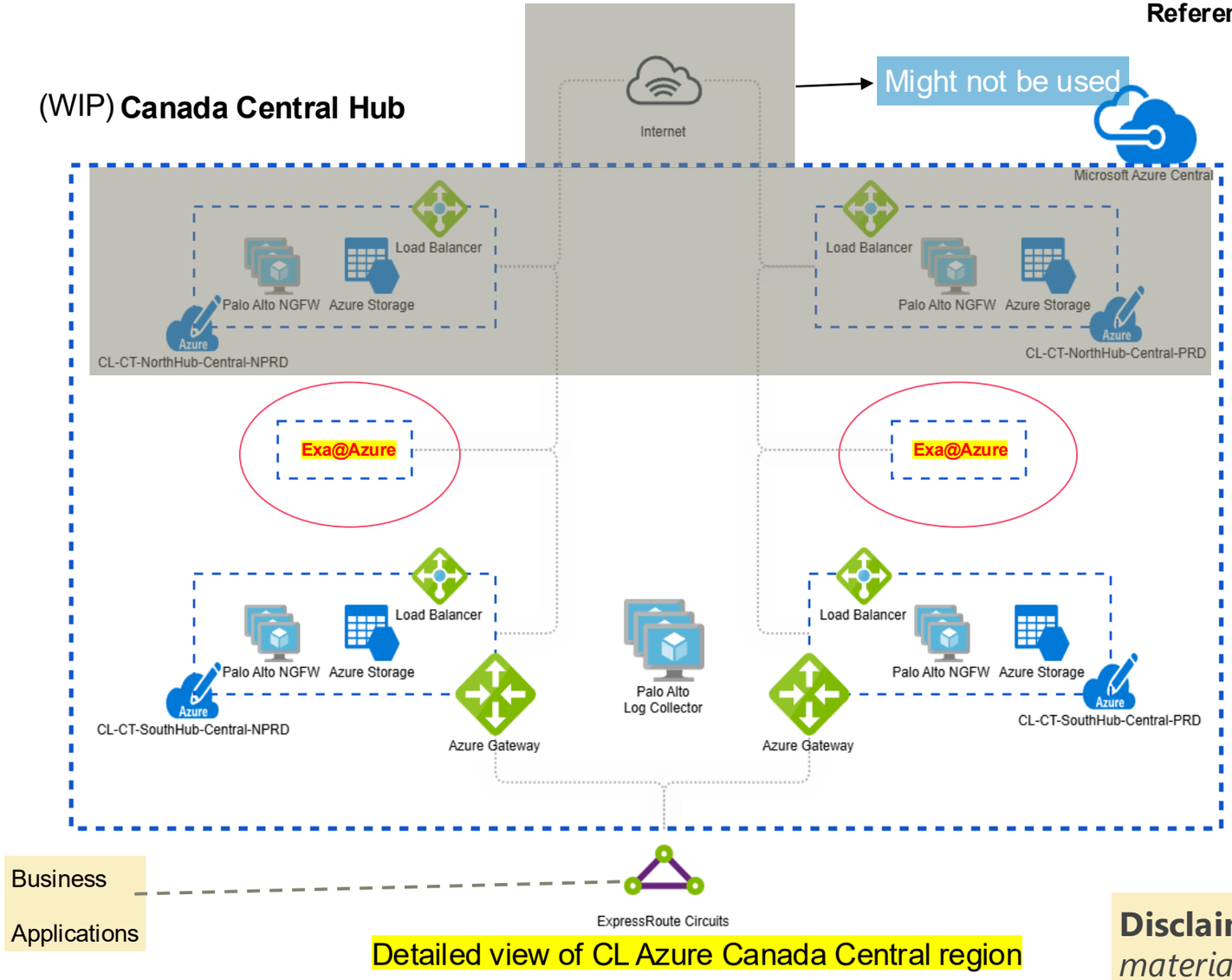
Conceptual Architecture Diagram (Systems Context)

Reference: AZ - Equinix Connections - Cloud & Automation - PRD-CONFLUENCE

(WIP) **Canada Central Hub**



Detailed view of CL Azure Canada Central region

Exadata on Azure is only supported in Canada Central region so both Prod and Non-Prod will be deployed here in Canada Central region.

Note: Oracle Cloud Infrastructure (OCI) currently provides only one Availability Domain in the Canada Southeast (Toronto) region. As a result, services offered by or that are directly related to OCI are subject to a slightly lower SLA of 99.95%.

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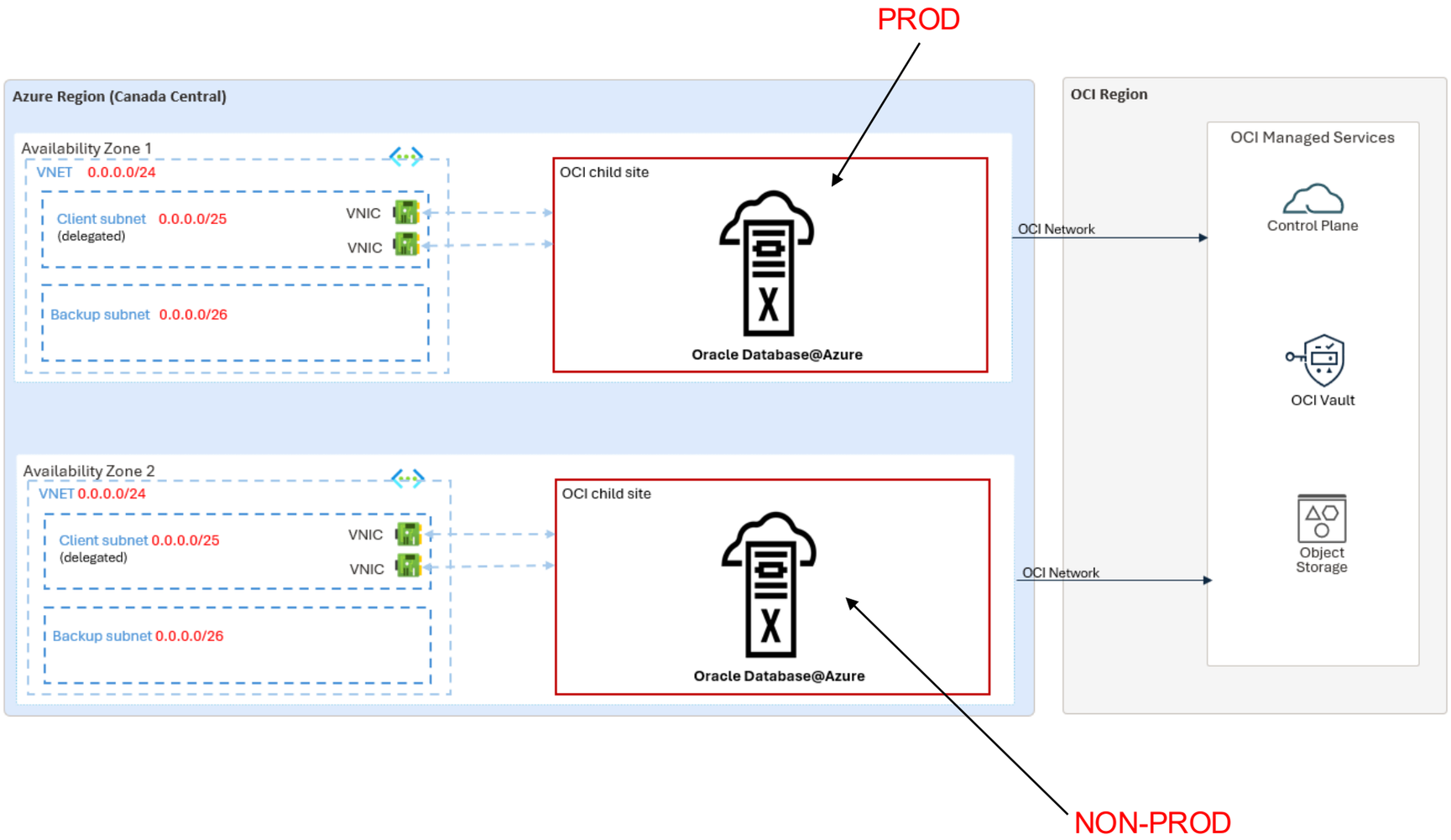
Azure Logical Architecture (Canada Central Region)



Summary:

1. This diagram illustrates how the Oracle Database on Azure is connected to the OCI network. Through this connection, Oracle provides Canada Life with a management interface to administer ExaDB instances. Additionally, Oracle can perform maintenance activities on ExaDB via the OCI network

2. Every Cluster of ExaData has a minimum of two nodes and each VNIC connects to a single node.

Second Exadata@azure appliance will be deployed in Canda East Region, once it is available and is not shown here.

the two production environments in Azure had to be deployed in separate regions due to compliance (OSFI) requirements.

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Additional Information

Oracle Database@Azure VM Cluster Configuration – In-region HA in a single Azure AZ using Oracle Real Application Clusters (RAC)

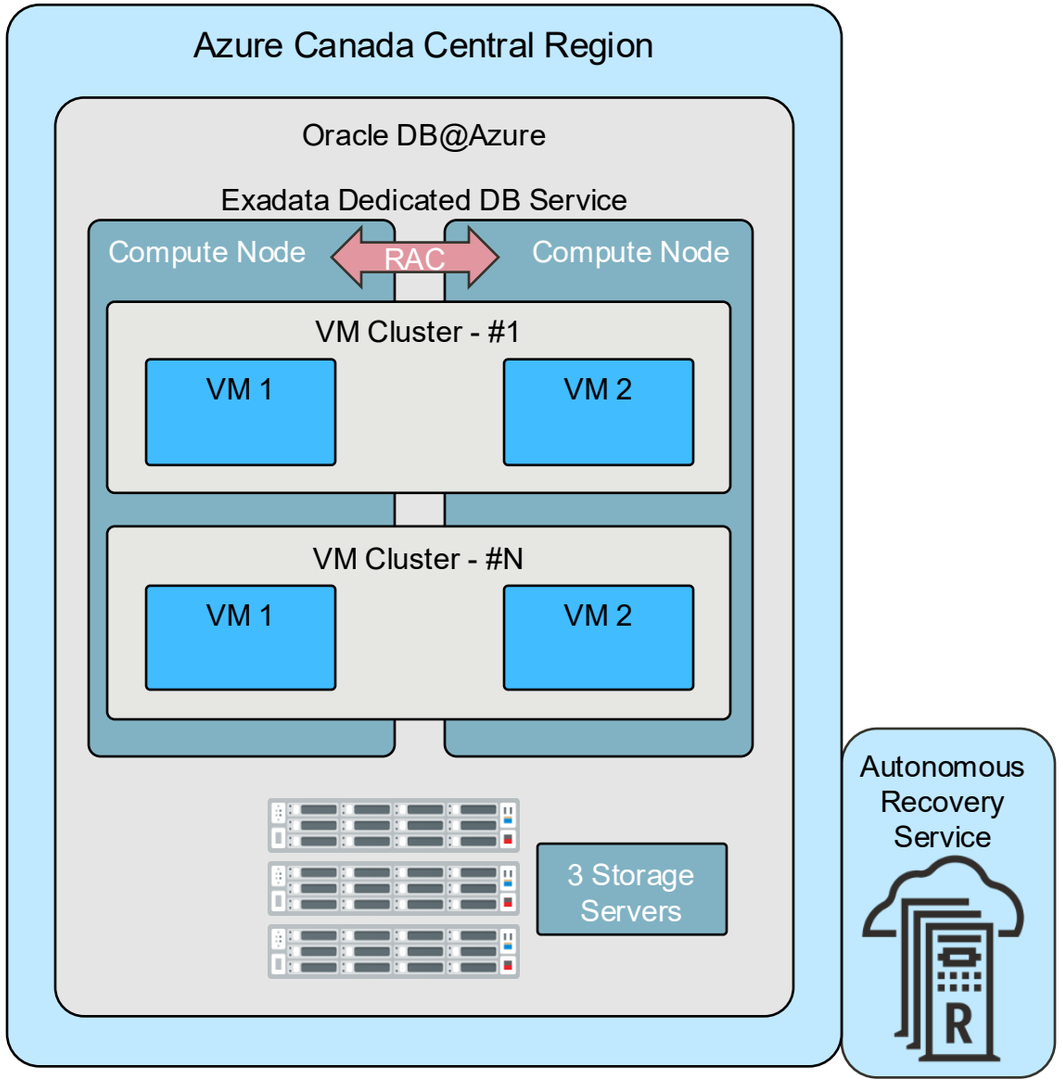

Image illustrates initial deployment mode for Canada Life in the Canada Central Region.

The specific details regarding the VM clusters and the virtual machines within them are yet to be finalized. However, the current inclination from CL is to deploy a single VM cluster for Workstream 1 in the non-production environment. This setup will be used to test system performance, which will inform the final configuration requirements for the production environment.

Future Consideration:

The **Autonomous Database** offering from Oracle Cloud Infrastructure (OCI) could be explored as a potential option. It offers the ability to automatically scale database instances based on actual usage patterns, which may lead to operational efficiency and potential cost savings for CL.
This option warrants further evaluation and research to assess its suitability and alignment with CL's long-term database strategy.

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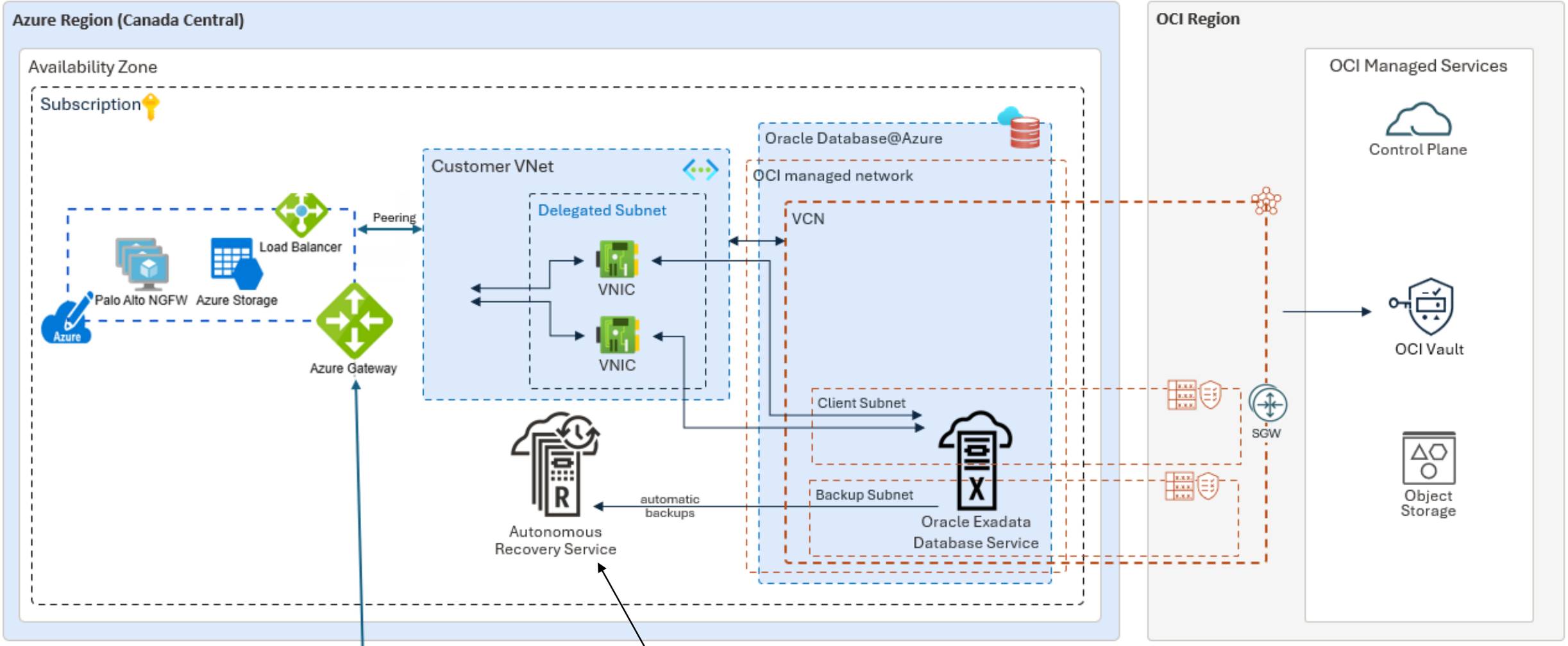
Physical Architecture

Oracle Maximum Availability Architecture for Oracle Database@Azure.



Business Applications

ExpressRoute Circuits

The backups (ARS) are replicated in the same region into a second availability zone (AZ)

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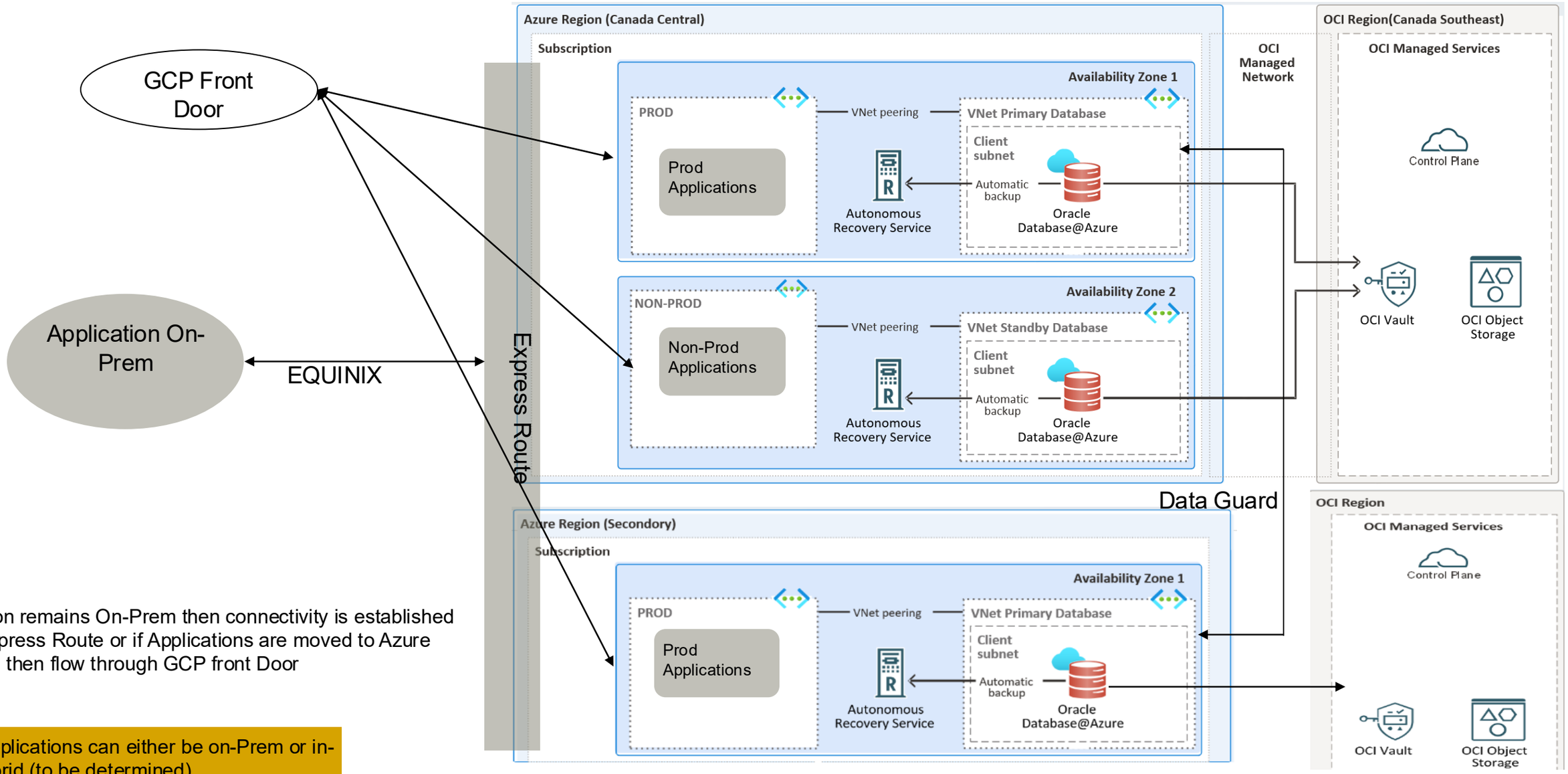
Physical Architecture (Future State when additional Azure Regions are available)

Oracle Maximum Availability Architecture for Oracle Database@Azure.

The backups are replicated in the same region into a second availability zone (AZ)



If Application remains On-Prem then connectivity is established through Express Route or if Applications are moved to Azure request will then flow through GCP front Door

In Future, Applications can either be on-Prem or in-Cloud or Hybrid (to be determined)

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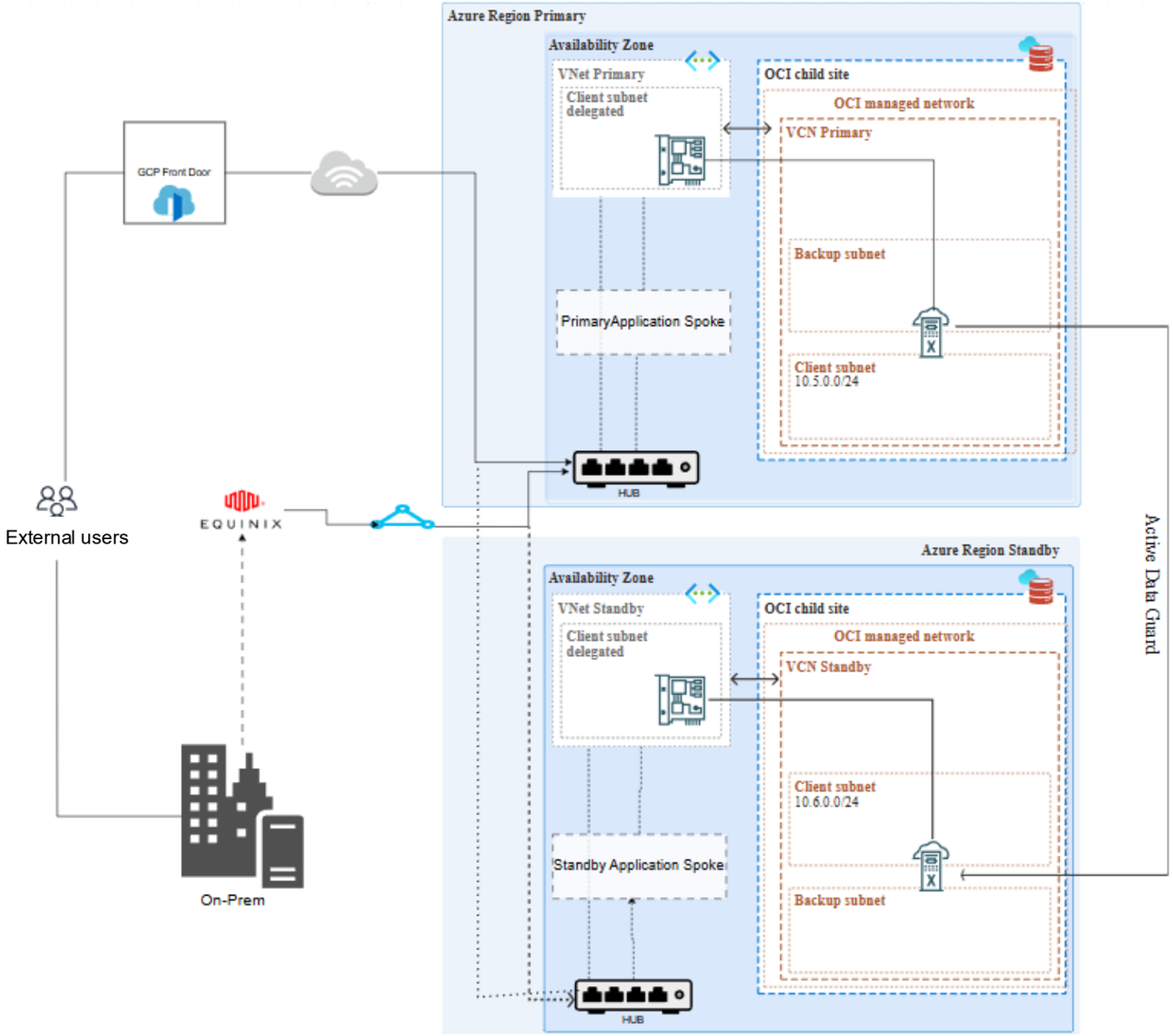
Future State – DR consideration



Decision Point for DR Strategy:

The DR setup depends on whether the applications remain on-prem or move to Azure. A final decision will be made by 30th November, following analysis of all 42 applications.

- If apps remain on-prem and Exadata is moved to Azure:
  - DR can be supported by hosting application servers in both SDC and WDC.
  - These servers will connect to Exa@Azure, which will be deployed in a multi-region setup.
  - Both active-active and active-passive strategies are under consideration.
- If apps move to Azure:
  - An active-active DR strategy is preferred for optimal resource use, though active-passive is still being evaluated.
  - In this case, GCP Frontdoor with Cloud Armour will route traffic to applications deployed in two Azure regions, each paired with its own Exa@Azure instance, synchronized via Oracle Active Data Guard

We may retain the current DR setup if the decision is made to upgrade the existing Exa@CC infrastructure.

Ref: GCP - Frontdoor Knowledge Base - Cloud & Automation - PRD-CONFLUENCE

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Important Links

S. No	Description	URL
1	Oracle Database Autonomous Recovery Service	Oracle Database Autonomous Recovery Service - Get Started
2	Network planning (Client Subnet)	Network planning for Oracle Database@Azure | Microsoft Learn

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Non-Functional Requirements

DBR Rating	Current resiliency class for individual applications is attached.
High Availability	Yes using 2 VMs within Exa@azure cluster, (Multi-region deployment to be considered for future stages)
Critical Operating Hours	24x7
Maintenance Windows	Midnight to 6am CT nightly (or negotiated across LoBs for certain activities). For Exa@Azure done by Oracle, CL can adjust the maintenance schedule as needed.
RTO	Listed in attached sheet for all 42 applications currently on Exa@CC
RPO	<= 24 (1 day)
What monitoring tools are leveraged?	Splunk, OCI,OEM and Trellix (Additional tool can be added as per need)
Does the NFT environment reflect production?	"no" with respect to Applications and "yes" with respect to platform itself.

Service Level Agreements	Green	Yellow	Red
% of transactions that are processed successfully	≥ 98%	< 98% ≥ 90%	< 90%

The attached sheet lists 42 Exa@CC applications with current key metrics.

Microsoft Excel Worksheet

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Non-Functional Requirements

The highest resiliency class among the 42 applications on Exa@CC is RC2. While Exa@azure with a single Availability Zone deployment offers a 99.95% SLA which corresponds to **RC3**, for future consideration a multi-region Azure setup can achieve 99.99% SLA.

Resiliency Classes

	RC1 Basic Availability (98.5%)	RC2 Moderate Availability (99%)	RC3 High Availability (99.9%)	RC4 Continuous Availability (99.999%)
Availability • Business Expectation	• 12X5X52 (7am-7pm EST) less holidays, negotiated maintenance windows	• 24X7X52 less negotiated maintenance windows	• 24X7X52 less negotiated maintenance windows	• 24X7X52
• Service Level Achievement (Outage Time per year)	• 60 business hours (5 business days)	• 88 hours (3.7 days)	• 9 hours (1.1 days)	• 5 minutes
Operational Recovery Time • P1/P2 Incident	< 60 business hours	< 4 hours	< 2 hours	< 5 minutes
Operational Recovery Point[1]	Last backup[2]	≤ 24 hours	≤ 24 hours	Seconds
DR RTO	Months	< 96 hours	< 24 hours	< 5 minutes
DR RPO[1]	Last backup[2]	≤ 24 hours	≤ 24 hours	Seconds
Design Considerations	• Many single points of failure • No recovery center capabilities • Basic instrumentation	• Minimal single points of failure • Single points of failure must be recoverable within 4 hours • Dedicated infrastructure at recovery center activated at time of disaster • Basic instrumentation	• In data center high availability with no single points of failure • Dedicated infrastructure at recovery center activated at time of disaster • Fully instrumented • Fully automated failover • Uninterrupted service during change	• Multi-site active operations with geographic high availability • Each site can handle a single down condition and has capacity to manage 100% work load • Fully instrumented • Fully automated failover • Uninterrupted service during change
Business Considerations	• Application is not transactional • Application is non-mission critical • There is a workaround for the application that can stay in place indefinitely • Pure testing environments may choose this class	• Application is internal but important • Application is batch transactional but not time sensitive • Extended outage can be tolerated	• The application is client facing but not transactional or at least not time sensitive • Application is highly visible but can tolerate minimal interruption	• The application is client-facing and transactional • There is no tolerance for downtime • Transactions are unscheduled and time sensitive
• Features	• Local continuity of services, with relatively seamless transition to backup gear	• Regulatory compliance • Investor confidence • General optics management in the event of outages • Isolated workload recovery	• Higher continuity of services driving positive CX – meeting general expectation of (apparently) "always on" services	• No down time required for patching/maintenance • Maximized asset utilization • Minimal legacy DR test program impact on organization

Notes:

[1] Foundational data centre services such as network fabrics where data is transient do not house data other than configurations and policies. RPOs do not necessarily apply to these services.

[2] RPO in basic availability class (RC1) is variable depending on requirements and availability of data protection services in a location. For example:

- ClickShare display services are site specific and would be rebuilt vs. restored at a business continuity site (no RPO).
- All systems in a data centre are backed up daily by default – applications in the basic availability class residing in a data centre have at minimum a ~24 hour RPO.

Appendix

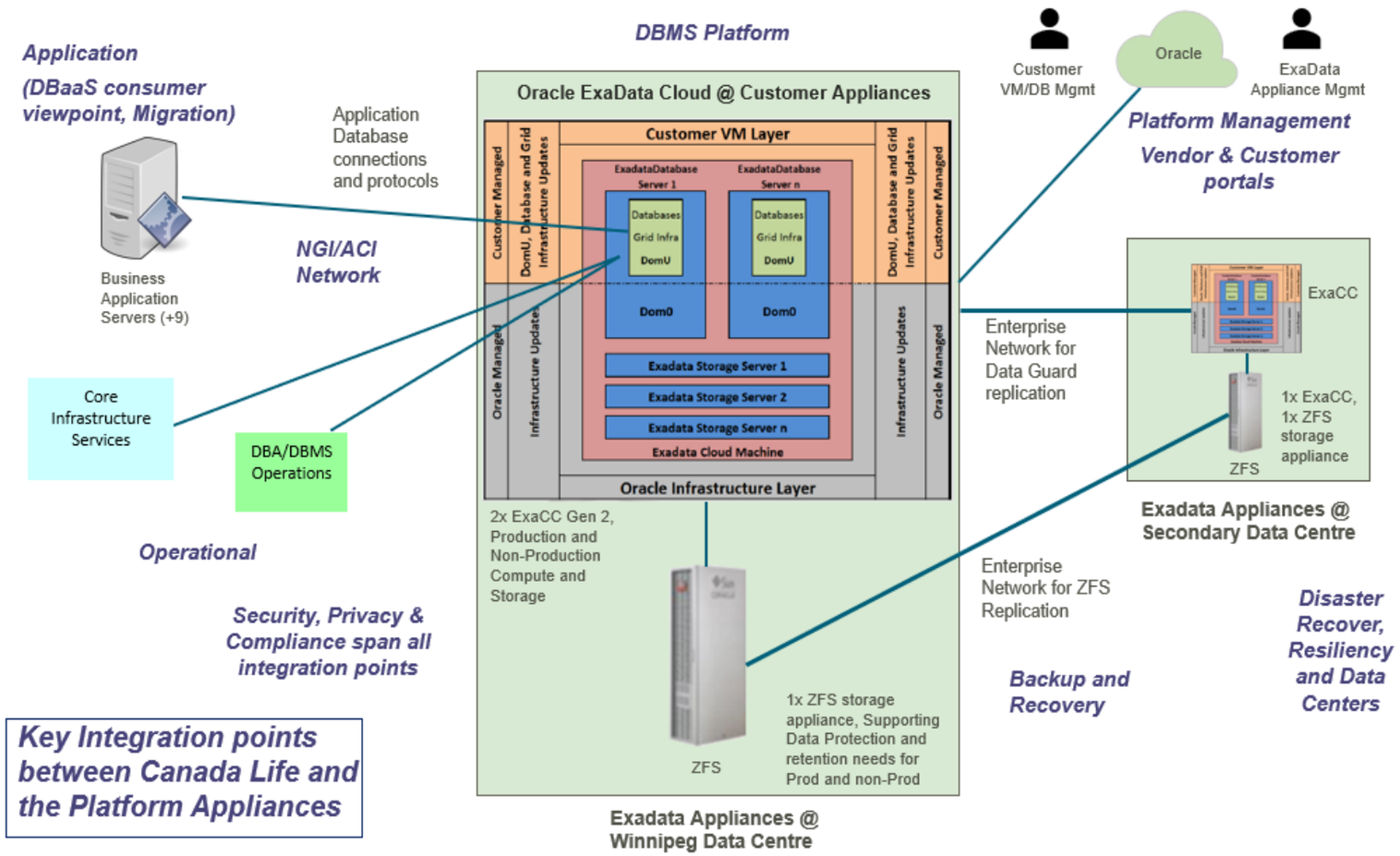
Systems Context

Summary:

The I&O Exadata in Azure initiative is a key component of Technology's broader modernization strategy aimed at leveraging cloud capabilities to enhance IT infrastructure and operational efficiency.

Currently, Oracle Exadata systems are hosted in Canada Life's WDC and SDC data centres. These systems run on Oracle-managed physical hardware that is approaching its mandatory refresh cycle. Canada Life has chosen to migrate Exadata workloads to the cloud. This strategic move aligns with the company's modernization goals, offering improved scalability, agility, and long-term cost efficiency.

Exadata Cloud @ Customer – Integration Perimeter



Exa@CC | CL managed layer | Oracle managed layer

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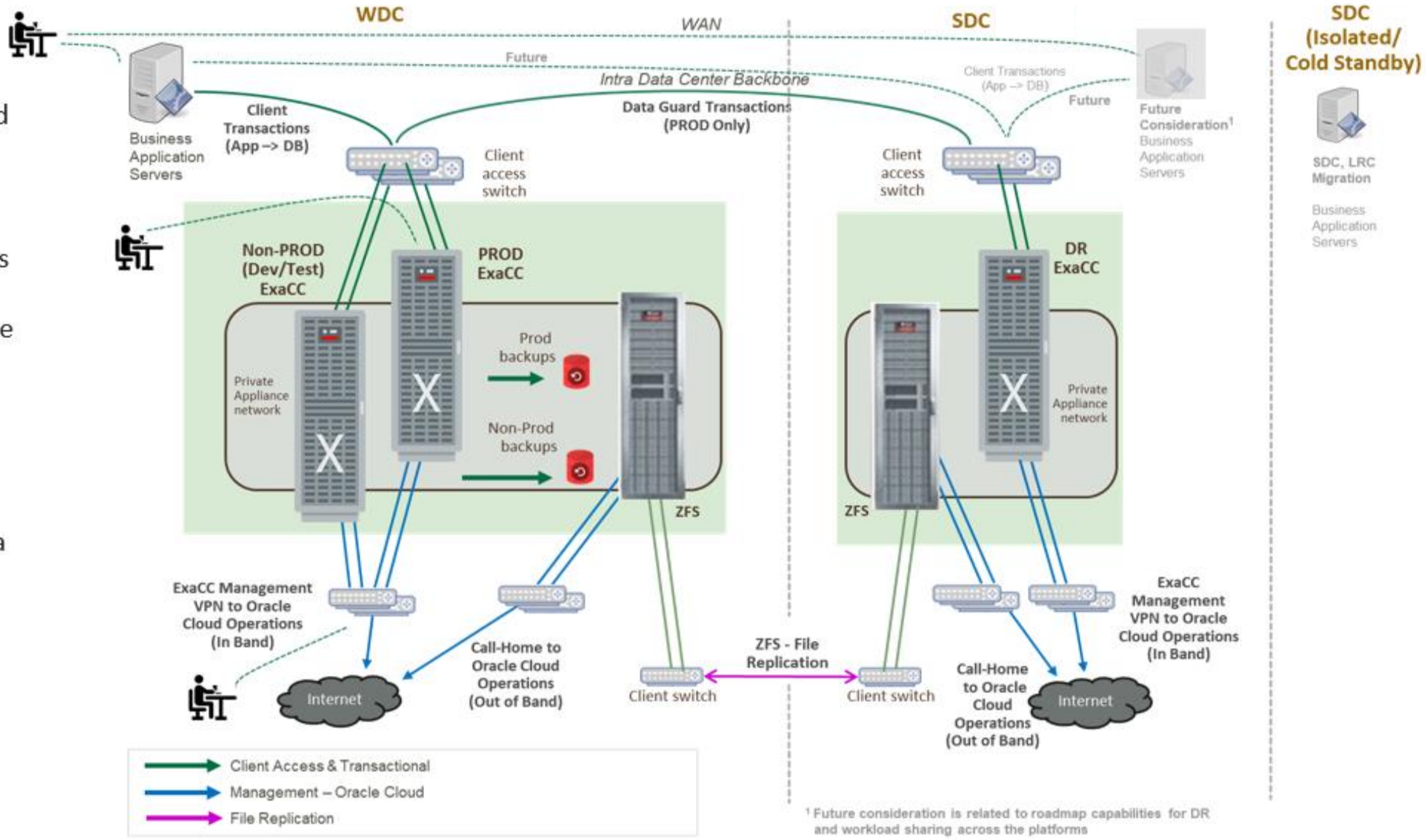
Physical Architecture

Technology View

Oracle platform is provisioned and managed by Oracle and resides in Canada life data centers.

Canada life applications subscribe to compute database services in the platform.

The platform provides cloud-level scalability and resilience integrated with Canada life systems.



[1] Future consideration is related to roadmap capabilities for DR and workload sharing across the platforms

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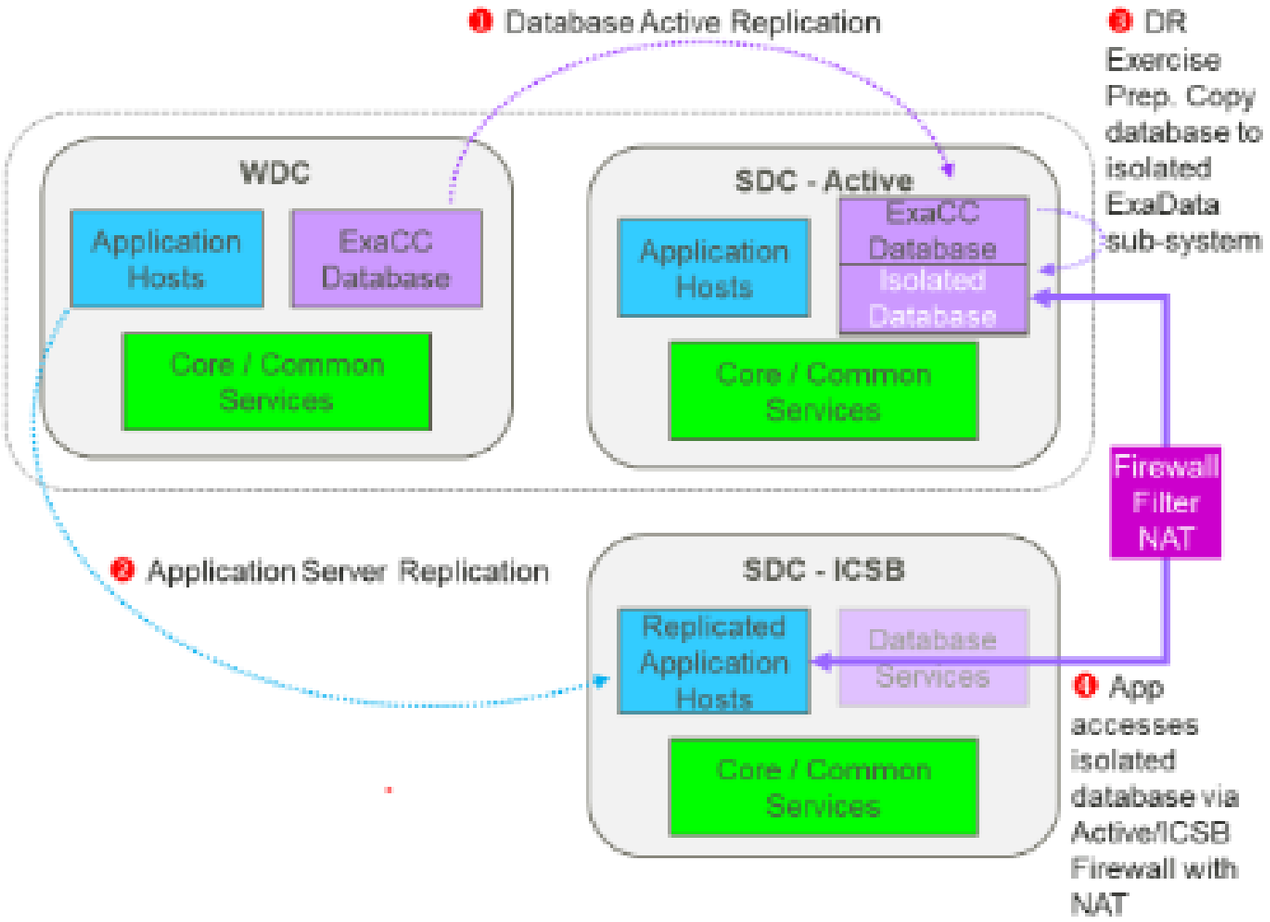
Disaster Recovery

Hybrid DR Exercise Pattern – ExaData Use Case

Traditional DR Exercises

- ❶ Database Active Replication
- ❷ Application Server Replication
- ❸ DR Exercise Preparation
- ❹ Application DR test in Isolated Data Center
  - Database connection configuration designed connect to multiple database instances (Primary/Secondary in ExaData ecosystem)
  - Application accesses ExaData isolated database through network connection to the SDC ExaData appliance via NAT or straddle patterns



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Additional Information

Azure Networking – ExaDB-D

Remember to pla

ExaDB-D – VM Cluster Networking in Azure

- Needs a VNet with 1x subnet (**Client subnet**) in Azure
  - Client subnet needs to be created and **delegated** to Oracle Database@Azure before ExaDB-D VM cluster deployment
- Need to provide **Backup subnet** CIDR block during VM cluster deployment
  - Backup subnet will be deployed in OCI, not in Azure
- Plan not to overlap network
  - Non overlapping CIDR blocks are needed for each deployment in a DR scenario

The following table shows the minimum required subnet size depending on ExaDB-D configuration you choose.
Each database node require 4x IP addresses and in addition 3x addresses are reserved for Single Client Access Names (SCANs.)

Rack Size	Client Subnet # Required IP Addresses	Client Subnet : Minimum size *	Backup Subnet # (Not in Azure) Required IP Addresses	Backup Subnet : (Not in Azure) Minimum size *
Flexible infrastructure (X9M and higher)	4 addresses per database node (minimum 2 nodes) + 3 for SCANs + 13 for ODBAA service	Minimum size determined by total number of IP addresses needed for database node	3 addresses per database node (minimum 2 nodes)	Minimum size determined by total number of IP addresses needed for database nodes

Reference :-

Size of subnet	Total number of IPs in subnet
/28	16
/27	32
/26	64
/25	128
/24	256
/23	512

Azure Networking for ExaDB-D – example

ExaDB-D Network examples : 2x or 4x Database nodes

	Client Subnet # IP Addresses in this example	Client Subnet : Minimum size in this example	Backup Subnet # (not in Azure)
Example 2x nodes	4x2 nodes + 3 = 11 IPs ODBAA requires 13 IPs Total = 24	/27 minimum	/28 minimum
Example 4x nodes	4x4 nodes + 3 = 19 IPs ODBAA requires 13 IPs* Total = 32	/26 minimum (recommend to to use larger for future expansions)	/27 minimum

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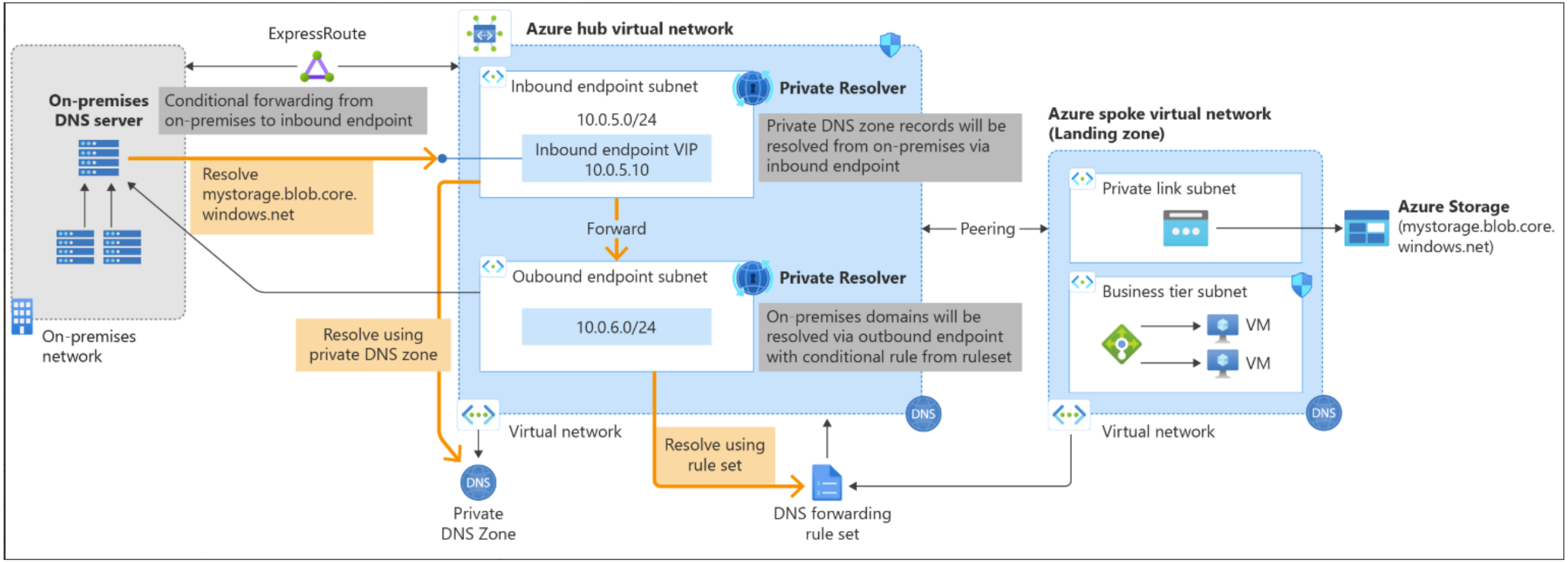
Additional Information (**DNS Private Resolver**)

Reference: AZU - DNS Private Resolver - Cloud & Automation - PRD-CONFLUENCE , What is Azure DNS Private Resolver? | Microsoft Learn



Disclaimer: *The image used above is sourced from the referenced material. It is assumed that the information depicted is accurate and exists as presented.*

Additional Information (**Data Migration**)

The decision regarding data migration from the on-premises non-production environment to Exa@Azure non-production is still pending. Currently, three options are under consideration:

1. Restore a backup taken from ExaCC to Exa@Azure.
2. Export the database, transfer it to Azure storage (e.g., BLOB), and then import it into Exa@Azure.
3. Leverage Data Guard for data replication.

These are preliminary options. A final decision will be made at a later stage, based on internal discussions and the availability of resources.

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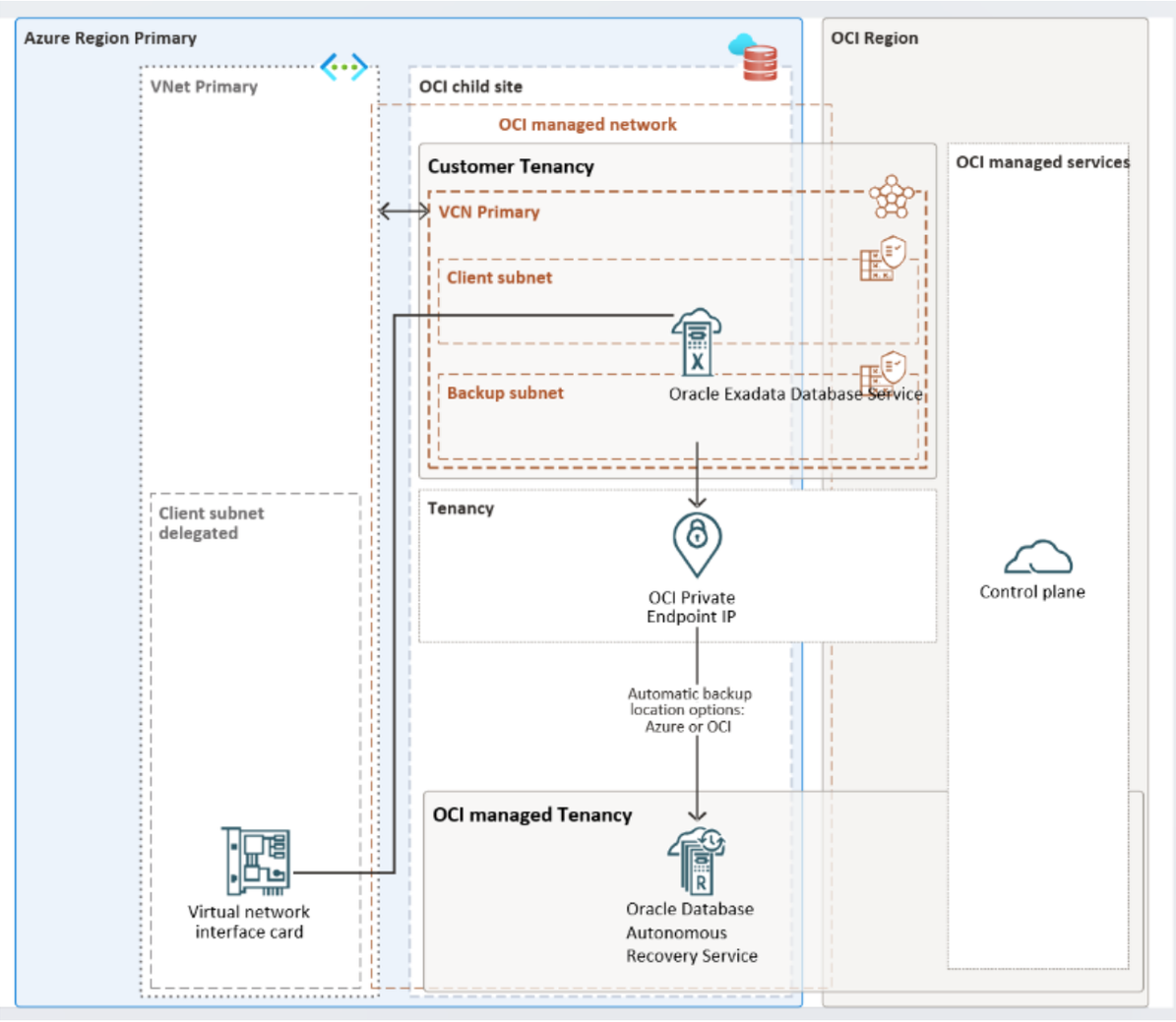
Physical Architecture (Autonomous recovery service- *Zero Data Loss Autonomous Recovery Service*)

Implement Oracle Database Autonomous Recovery Service with Oracle Database@Azure



In a multicloud deployment, the backups are replicated in the same region into a second availability zone (AZ), or fault domain (if another AZ is not available) in Azure

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Oracle Database Service for Azure Architecture

Reference: Oracle Database Service for Microsoft Azure Technical Overview

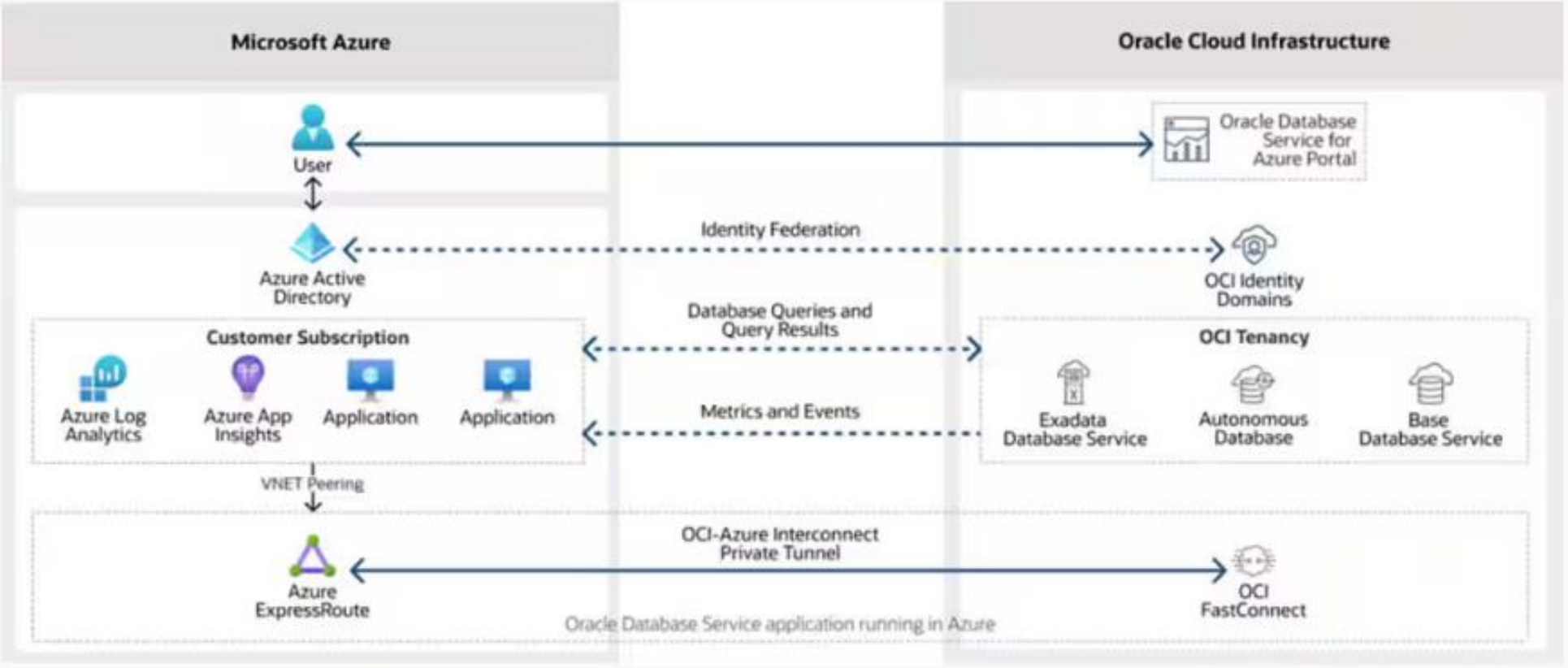

Figure 2 – Oracle Database Service for Azure Architecture

Note: OCI and Azure are interconnected via the Azure Interconnect, as illustrated in the next slide. While the Canada Central region of Azure offers three Availability Zones, enabling a multi-zone deployment that supports an SLA of 99.99%, Oracle Cloud Infrastructure (OCI) currently provides only one Availability Domain in the Canada Southeast (Toronto) region. As a result, services deployed or linked to OCI are subject to a slightly lower SLA of 99.95%.